DOCUMENT: Bengaluru Hyperscale Data Centre - UPS Technical Specification (R01)
SECTION 7.4: UPS EFFICIENCY REQUIREMENTS

1. Executive Summary: This specification defines technical parameters for all Uninterruptible Power Supply (UPS) units.
2. Efficiency Baseline: The minimum required UPS efficiency is 96% across operating loads from 50% to 100%.
3. High Efficiency Mode: Equipment capable of eco-mode operation must still maintain a double-conversion efficiency of at least 96%.
4. Heat Rejection: Thermal dissipation calculations must be derived based on the 96% efficiency baseline.

DOCUMENT: Bengaluru Hyperscale Data Centre - UPS Technical Specification (R01)
SECTION 7.5: BATTERY STORAGE AND AUTONOMY

1. Battery Autonomy Requirement: The battery system shall provide a minimum of 15 minutes of autonomy at full rated 48 MW load.
2. Cell Chemistry: Lithium-ion battery racks rated for fast discharge capability.
3. Runtime Compliance: Submittals featuring autonomy below 15 minutes shall be rejected as non-compliant.

DOCUMENT: Bengaluru Hyperscale Data Centre - UPS Technical Specification (R01)
SECTION 7.6: SYSTEM REDUNDANCY AND TOPOLOGY

1. Redundancy Configuration: The UPS system topology shall be configured for N+1 redundancy.
2. Fault Isolation: Each parallel UPS module shall include independent static bypass switches and isolation breakers.
3. Concurrent Maintainability: System topology must ensure full concurrent maintainability without taking down IT loads.

DOCUMENT: Bengaluru Hyperscale Data Centre - UPS Technical Specification (R01)
SECTION 8.1: EMERGENCY GENERATOR SYSTEMS

1. Fuel Autonomy Baseline: Generator diesel fuel storage capacity must guarantee a minimum generator runtime of 48 hours.
2. Fuel Transfer System: Dual redundant transfer pumps with day tank automatic fill capability.
3. Continuous Rating: Generators must be rated for continuous mission-critical standby service.

DOCUMENT: Bengaluru Hyperscale Data Centre - UPS Technical Specification (R01)
SECTION 9.2: COOLING SYSTEM SPECIFICATIONS

1. Cooling Redundancy: Data hall cooling system configuration must maintain N+1 redundancy for all chiller units and CRAH modules.
2. Temperature Setpoints: Supply air temperature to be maintained at 24°C ± 1°C.
3. Chilled Water Loop: Dual ring header topology for pipework fault isolation.